

# CYBER-PHYSICAL SYSTEMS MINOR

## Cyber-Physical Systems Minor

With the fourth industrial revolution upon us, physical systems are being designed to have a cyber component, that enables remote access, monitoring and control. In these systems, ubiquitous sensing, and advanced data management capability are taking us from automation to autonomy via a deep interconnection between the cyber and physical entities. Cyber-physical systems (CPS) are becoming abundant in many application sectors including manufacturing, energy, health care, transportation and agriculture. Safety-, time- and life-critical systems are relying on CPS concepts to become more efficient, robust, resilient, flexible and scalable. As CPS applications become more pervasive, the engineering education system needs to produce a next generation CPS workforce who can design, produce, and maintain these systems.

With this motivation and encouraged by the demand from the industry stakeholders of ISU College of Engineering, this CPS minor will focus on sensing, advanced information processing (data analytics and machine learning), and controls aspects of Cyber-Physical Systems. Specific CPS application sectors such as energy/power systems, manufacturing, biomedical devices, autonomous systems, transportation, and agriculture will be in focus.

The minor requires 15 credits, including at least 6 credits in courses numbered 3000 or above taken at Iowa State University. The minor must include 3 credits that are not used to meet any other department, college, or university requirement.

Required courses:

|           |                                               |   |
|-----------|-----------------------------------------------|---|
| ME 2800   | Design and Analysis of Cyber-Physical Systems | 3 |
| AERE 3640 | Cyber-Physical Systems Application            | 3 |
| CPRE 2870 | Cyber-Physical System Fundamentals            | 3 |

Elective courses: 6

|           |                                                   |
|-----------|---------------------------------------------------|
| ME 3700   | Engineering Measurements                          |
| ME 4110   | Automatic Controls                                |
| ME 4180   | Mechanical Considerations in Robotics             |
| ME 4210   | System Dynamics and Control                       |
| ME 4560   | Machine Vision                                    |
| ME 4750   | Modeling and Simulation                           |
| EE 3240   | Signals and Systems II                            |
| EE 3330   | Electronic Systems Design                         |
| EE 4250   | Machine learning: A Signal Processing Perspective |
| EE 4760   | Control System Simulation                         |
| CPRE 2300 | Cyber Security Fundamentals                       |

|               |                                                             |    |
|---------------|-------------------------------------------------------------|----|
| CPRE 3880     | Mobile Platforms and Applications                           |    |
| CPRE 4140     | Introduction to Software Systems for Big Data Analytics     |    |
| CPRE 4190     | Software Tools for Large Scale Data Analysis                |    |
| CPRE 4210     | Software Analysis and Verification for Safety and Security  |    |
| CPRE 4580     | Real Time Systems                                           |    |
| CPRE 4880     | Embedded Systems Design                                     |    |
| ABE 4030      | Physical Modeling and Control of Dynamic Systems            |    |
| ABE 4040      | Instrumentation for Agricultural and Biosystems Engineering |    |
| ABE 4100      | Electronic Systems Integration for Agricultural Machinery   |    |
| IE 4130       | Stochastic Modeling, Analysis and Simulation                |    |
| IE 4320       | Industrial Automation                                       |    |
| IE 4870       | Big Data Analytics and Optimization                         |    |
| AERE 4070     | Applied Formal Methods                                      |    |
| AERE 4330     | Spacecraft Dynamics and Control                             |    |
| AERE 4630     | Introduction to Multidisciplinary Design Optimization       |    |
| AERE 4640     | Spacecraft Systems                                          |    |
| CE 4490       | Structural Health Monitoring                                |    |
| CE 4530       | Highway Design                                              |    |
| CE 5530       | Traffic Engineering                                         |    |
| CE 5560       | Transportation Data Analysis                                |    |
| Total Credits |                                                             | 15 |